

Question	Answer	Marks
9(a)	emission of electrons (from a metal surface)	M1
	when electromagnetic radiation is incident (on surface)	A1
9(b)(i)	threshold frequency = $2F$	A1
9(b)(ii)	Planck constant = ϕ / F	A1
9(c)	two diagonal straight lines with positive gradient in the positive E_k region only, starting at non-zero values of f	B1
	line labelled X passing through $(F, 0)$ and line labelled Y passing through $(2F, 0)$ and extending to $3F$ or $+2\phi$	B1
	two diagonal straight lines with equal gradients	B1
	X line extrapolates back to $(0, -\phi)$ and Y line extrapolates back to $(0, -2\phi)$	B1

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